OBLIG 2

$$f(x) = e^{-\frac{x}{4}} \cdot \tan^{-1} x$$

Vis at maksimal punktet er vist med:

$$\tan^{-1} x - \frac{4}{x^2+1} = 0$$

$f'(x) =$

$u = e^{-\frac{x}{4}}$

$v = \tan^{-1} x$

$u' = e^{-\frac{x}{4}} \cdot \left(-\frac{1}{4}\right)$

$v' = \frac{1}{1+x^2}$

$u' \cdot v + u \cdot v'$

$$e^{-\frac{x}{4}} \cdot \left(-\frac{1}{4}\right) \cdot \tan^{-1} x + e^{-\frac{x}{4}} \cdot \frac{1}{1+x^2}$$

$$-\frac{e^{-\frac{x}{4}} \cdot \tan^{-1} x}{4} + \frac{e^{-\frac{x}{4}}}{1+x^2} = 0$$

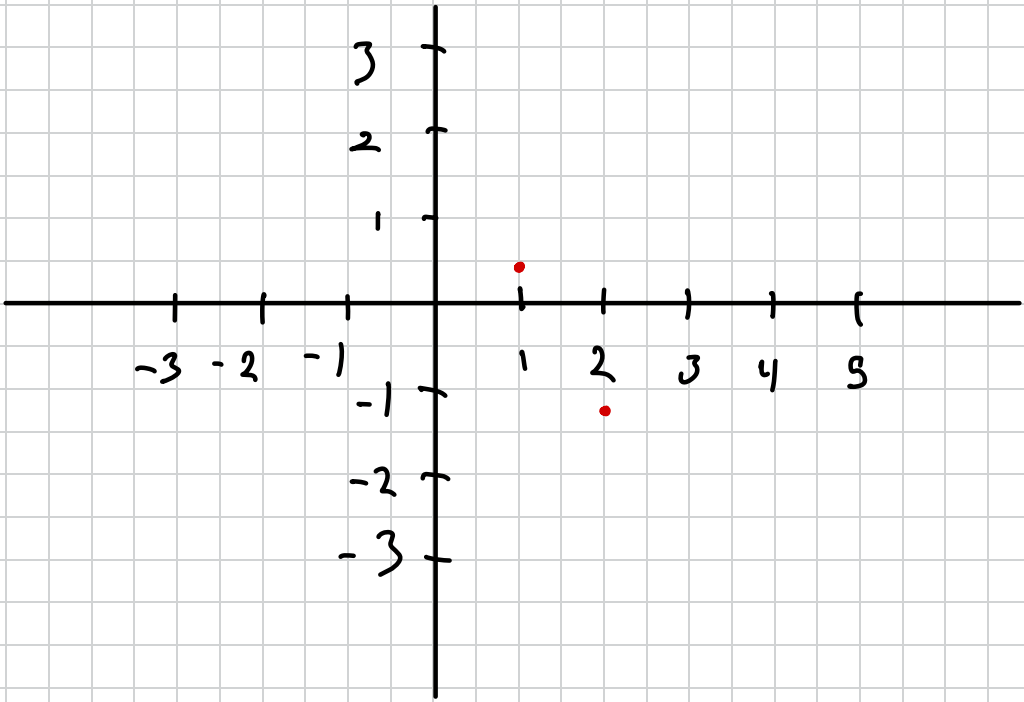
$$\frac{e^{-\frac{x}{4}}}{1+x^2} = \frac{e^{-\frac{x}{4}} \cdot \tan^{-1} x}{4} \quad | : e^{-\frac{x}{4}}$$

$$\frac{1}{1+x^2} = \frac{1 \cdot \tan^{-1} x}{4}$$

$$\frac{1}{x^2+1} = \frac{\tan^{-1} x}{\cancel{4}} \quad | \cdot 4$$

$$\underline{-\tan^{-1} x + \frac{4}{x^2+1} = 0} = \underline{\tan^{-1} x - \frac{4}{x^2+1} = 0}$$

$$\tan^{-1} x = \frac{4}{x^2+1}$$ | Tester verdier



$x = 1 \rightarrow \tan^{-1}(1) - \frac{4}{1^2+1} = -1.21460$

$x = 2$ —— " —— $= 0.30714$

$x = 1.1$ —— $> \tan^{-1}(1.1) - \frac{4}{1.1^2+1} =$

$x = 1.2$ $\tan^{-1}(1.2) - \frac{4}{1.2^2+1} = -1.071780638$

$x = 1.3$ $\tan^{-1}(1.3) - \frac{4}{1.3^2+1} =$

$x = 1.4$ $\tan^{-1}(1.4) - \frac{4}{1.4^2+1} =$

$x = 1.5$ $\tan^{-1}(1.5) - \frac{4}{1.5^2+1} =$

$x = 1.6$ $\tan^{-1}(1.6) - \frac{4}{1.6^2+1} =$

$x = 1.7$ $\tan^{-1}(1.7) - \frac{4}{1.7^2+1} =$

testet

Ligger mellom

✓ 1.688 og 1.689

$x = 1.6907 \Rightarrow \tan^{-1}(x) - \frac{4}{x^2+1} = \underline{-0.000009522894}$

$\approx (1.6907, 1.5820)$

$f(1.6907) = e^{\frac{1.6907}{4}} \cdot \tan^{-1}(1.6907) = 1.582000$